ECA 1011 – MICROPROCESSOR AND MICROCONTROLLERS

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Experiment 03: Multi-Byte Data and Array Operations Using 8086 Indirect Addressing

Objectives

1. To add two numbers using register-indirect and based addressing modes.
2. To find the sum of N 16-bit numbers stored in a word array.
3. To add two 64-bit numbers byte by byte with carry propagation using a loop.

Components Required

- 8086 Emulator / Simulator (e.g., emu8086) or 8086 Microprocessor Kit

Procedure

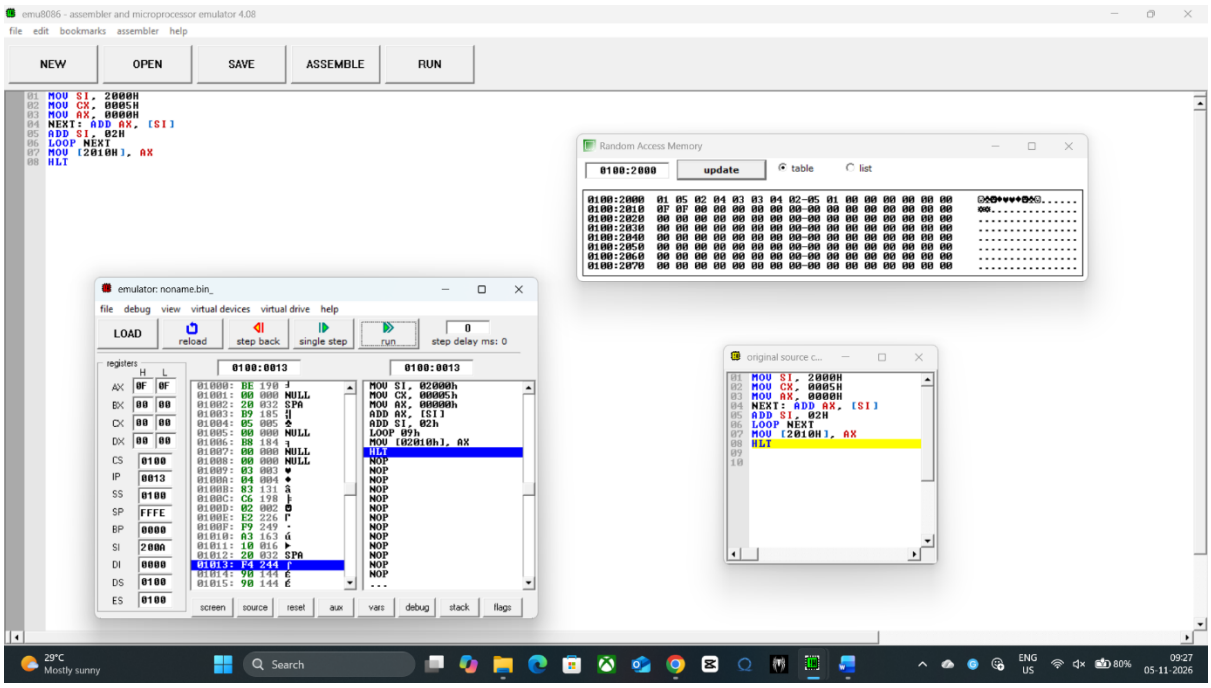
1. Launch your 8086 Emulator/Simulator application on your system.
2. Type or paste the Assembly Language Program (ALP) into the editor window.
3. Assemble/Compile the code to verify that there are no syntax errors or invalid segments.
4. Open the Memory Window and manually enter your input parameters or array bytes at the targeted data segment offsets.
5. Execute the program using Single Step (F7) or Run to track register movements, pointer increments, and flag changes (Carry, Zero, etc.).
6. Verify your final multi-byte calculations or array totals by checking the specified destination memory offset addresses.

2. **To find the sum of N 16-bit numbers stored in a word array.Code:**

Sum of N numbers:

Code:

```
MOV SI, 2000H
MOV CX, 0005H
MOV AX, 0000H
NEXT: ADD AX, [SI]
ADD SI, 02H
LOOP NEXT
MOV [2010H], AX
HLT
```



Data Table:

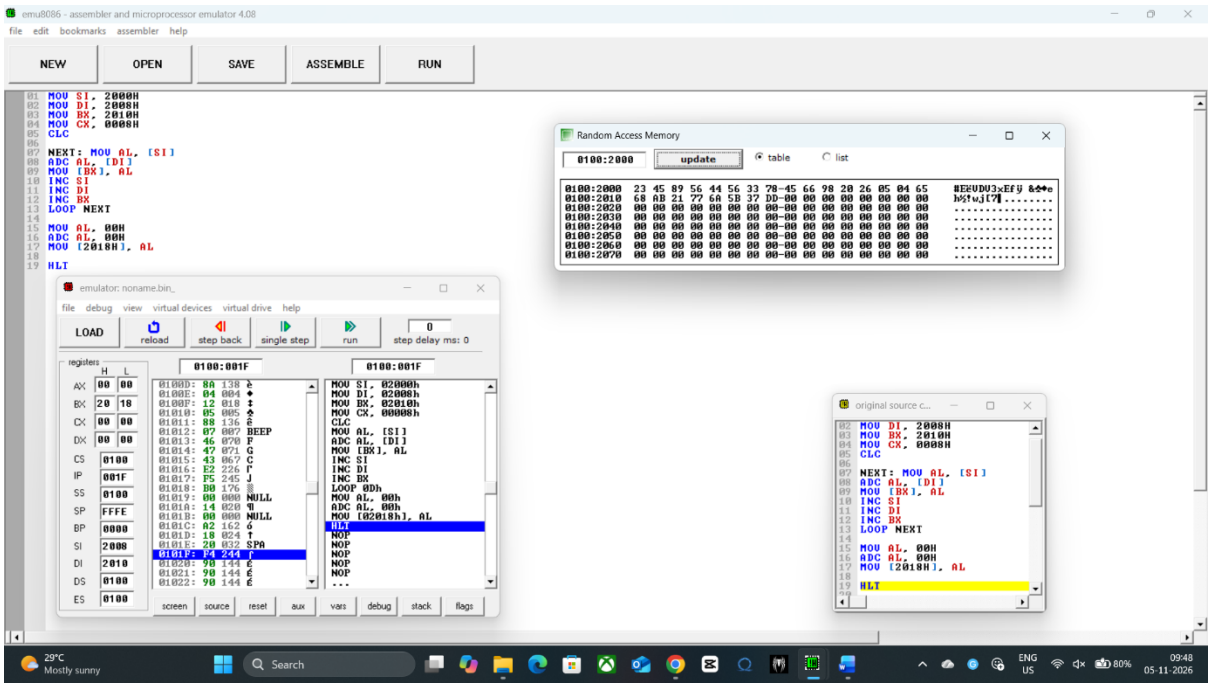
2000	**0105**
2002	**0204**
2004	**0303**
2006	**0402**
2008	**0501**
2010	**0F0F**

3. **To add two 64-bit numbers byte by byte with carry propagation using a loop.**

Code:

64-bit Addition:

```
MOV SI, 2000H
MOV DI, 2008H
MOV BX, 2010H
MOV CX, 0008H
CLC
NEXT: MOV AL, [SI]
ADC AL, [DI]
MOV [BX], AL
INC SI
INC DI
INC BX
LOOP NEXT
MOV AL, 00H
ADC AL, 00H
MOV [2018H], AL
HLT
```



Data Table:

2000	**23**
2001	**45**
2002	**89**
2003	**56**
2004	**44**
2005	**56**
2006	**33**
2007	**78**
2008	**45**
2009	**66**
200A	**98**
200B	**20**
200C	**26**
200D	**05**
200E	**04**
200F	**65**
2010	**68**
2011	**AB**
2012	**21**
2013	**77**
2014	**6A**
2015	**5B**
2016	**37**
2017	**DD**

Result:

1. **Addition and Subtraction using Register-Indirect and Based Addressing:** The target memory location SUM securely holds the correct 16-bit sum of 69ACH. The terminal flags accurately indicate that no overflow carry or borrow occurred during the operation.
2. **Sum of N 16-bit Numbers Stored in a Word Array:** The sequential destination memory address assigned to TOTAL contains the accurate, accumulated 16-bit array sum of FFF5H. The loop counter register CX successfully decremented to 0000H upon completion.
3. **Addition of Two 64-Bit Numbers Byte by Byte with Carry Propagation:** The designated 8-byte target memory block RESULT accurately holds the multi-byte product generated by the carry-propagating loop. Each index byte from LSB to MSB successfully absorbs and passes the preceding carry states.